

LAB WORKSHEET

Lab 32 — Network Connectivity Troubleshooting

Name		Date	
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Objective: 4.5 — Troubleshoot network connectivity: DNS/DHCP failure, routing, VLAN, port, cabling.
Work through the lab on the Killercoda Ubuntu playground and record your results below.

Task 1 — Layer 1-2

Check link state and ARP.
Record Link detected and the gateway neigh state.

Task 2 — Routing

Inspect routes and route-get a public IP.
Record which gateway is selected.

Task 3 — DNS chain

dig / nslookup a name and check hosts overrides.
Map one symptom to its cause.

Task 4 — Port test

Test a port with nc and scan with nmap.
Record open vs. filtered results.

Task 5 — Decision tree

Apply the bottom-up decision tree.
Write the eight-step order.

Completion checklist

- Checked link state and ARP at layers 1-2.
- Inspected routing and reachability.
- Diagnosed the DNS chain.
- Tested ports and captured packets.
- Applied the bottom-up decision tree.